around and vibrating, all with their endpoints restricted to motion along their respective branes. Although it is increasingly difficult to draw, the situation with higher-dimensional branes is identical. Open string endpoints can move freely on and within the *p*-brane, but they can't leave the brane itself. When it comes to the possibility of motion off a brane, branes are the stickiest things imaginable. It's also possible for one end of an open string to be stuck to one *p*-brane and its other end to be stuck to a different *p*-brane, one that may have the same dimension as the first (Figure 13.2b) or may not (Figure 13.2c).

To Witten's discovery of the connection between the various string theories, Polchinski's new paper provided a companion manifesto for the second superstring revolution. While some of the great minds of twentieth-century theoretical physics had struggled and failed to formulate a theory containing fundamental ingredients with more dimensions than dots (zero dimensions) or strings (one dimension), the results of Witten and Polchinski, together with important insights of many of today's leading researchers, revealed the path to progress. Not only did these physicists establish that string/M-theory contains higher-dimensional ingredients, but Polchinski's insights in particular provided a means for analyzing their detailed physical properties theoretically (should they prove to exist). The properties of a brane, Polchinski argued, are to a large extent captured by the properties of the vibrating open strings whose endpoints it contains. Just as you can learn a lot about a carpet by running your hand through its pile—the snippets of wool whose endpoints are attached to the carpet backing—many qualities of a brane can be determined by studying the strings whose endpoints it clutches.

